

Sukkeun KIM, PhD

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Bio

Sukkeun Kim is an aerospace engineer specialising in nonlinear Bayesian filtering and navigation systems. He bridges the gap between advanced mathematical theory and real-world implementation by developing and testing multiple estimation algorithms including particle flow-based and Gaussian mixture filters. He earned his BSc and MSc from Pusan National University and his PhD from Cranfield University. His research experience includes roles as a visiting student researcher at NASA's Jet Propulsion Laboratory (JPL) and the KAIST Graduate School of Mobility, focusing on applications in multi-target tracking, autonomous robotic localisation and space navigation systems.

SKILLS AND INTERESTS

Technical Expertise: Navigation/Localisation, State Estimation, Nonlinear Filtering (EKF, UKF, PF, GMF, PFF), Multi-Target Tracking (JPDA), Sensor Fusion, Computer Vision, SLAM

Languages & Tools: C++, Python, MATLAB/Simulink, Julia, Linux Ubuntu, ROS 1/2, OpenCV, Jetson TX2/Nano

TECHNICAL AND RESEARCH EXPERIENCE

Intelligent Sensor-Actuator-Systems (ISAS), KIT

Postdoctoral Researcher (Wissenschaftlicher Mitarbeiter)

Karlsruhe, Germany

September 2025–September 2026

- **Nonlinear Filtering:** Developed advanced particle flow-based filters (PFFs) for applications to nonlinear systems
- **Localisation Curriculum Design:** Appointed to design and lecture a full-semester master's course on *Localization of Mobile Agents (LMA)*, managing technical transfer of Bayesian estimation, sensor fusion and localisation frameworks

CCS Graduate School of Mobility, KAIST

Visiting Researcher

Daejeon, Republic of Korea

February 2025–August 2025

- PhD thesis project and supervision of graduate school students

NASA Jet Propulsion Laboratory (JPL)

JPL Visiting Student Research Program (JVS RP)

Pasadena, USA

September 2024–February 2025

- **Autonomous Space Navigation Pipelines:** Developed and evaluated advanced Bayesian estimation for autonomous robotic signal source localisation in unknown and unstructured environment
- **Algorithm Frameworks:** Implemented radio signal source localisation architecture on Linux Ubuntu and ROS 1

Cranfield University

PhD Researcher & Part-time Research Assistant

Cranfield, UK

September 2021–January 2026

- **Nonlinear Filtering Architecture:** Designed, simulated and deployed hybrid Bayesian estimation architectures combined with AI approaches for target tracking and source localisation in highly nonlinear and complex environments
- **Algorithm Development:** Programmed modular state-estimation pipelines for nonlinear Bayesian estimation and vision-based multi-class, multi-object tracking frameworks based on CNN and JPDA
- **Technical Mentor:** Guided laboratory implementation sessions using MATLAB, instructing students on practical sensor fusion module integration and Master's thesis supervision
- **RA Project (Part-time, October 2023–August 2024):** Conducted a study on operational procedures for UAM/AAM take-off and landing at vertiports adjacent to airports

NEXTFOAM

Research Intern

Seoul, Republic of Korea

March 2021–July 2021

- **Vision Algorithm:** Developed and evaluated a vision-based tracking algorithm and stereo vision system
- **Embedded Systems:** Implemented the architecture C++ on embedded system (Jetson Nano with camera)

Pusan National University

Part-time Research Assistant

Busan, Republic of Korea

August 2019–December 2020

- **Embedded Flight Software:** Engineered vision-based relative position measurement and tracking software in C++ deployed directly onto embedded NVIDIA Jetson TX2 hardware
- **Verification & Validation (V&V):** Successfully completed hardware-in-the-loop validation, collaborative UAV-UGV team field operations and live flight test verification
- **Relative Navigation Systems:** Designed deep-learning-based relative navigation software validated via light sport aircraft flight test data and collaboration with research institutes

EDUCATION

Cranfield University <i>PhD in Aerospace, Faculty of Engineering and Applied Sciences</i> - Thesis (Supervisors: Professor Hyo-Sang Shin and Professor Ivan Petrunin): Hybrid Bayesian Estimation Approaches for Tracking and Localisation in Complex Environments <i>Visiting Student, Centre for Autonomous and Cyber-Physical Systems</i> - Autonomous Vehicles Control System - Guidance and Navigation for Autonomous Systems	Cranfield, UK January 2026 December 2015–February 2016
Pusan National University <i>MSc in Aerospace Engineering (CGPA: 4.19/4.5)</i> - Thesis (Supervisor: Professor Daewoo Lee): Vision-based Relative Navigation with CNN-aided 3D Model Matching for Non-cooperative Airplanes <i>BSc in Aerospace Engineering (CGPA: 3.77/4.5)</i> - Thesis (Supervisor: Professor Daewoo Lee): Relative Position/Attitude Measurement of Leader Aircraft in Formation Flight using MBT	Busan, Republic of Korea February 2021 February 2019

SELECTED PEER-REVIEWED PUBLICATIONS (FULL LIST:)

Out of 6 Journals, 2 Preprints, 5 Conferences including IEEE, Information Fusion and IFAC - P2: S. Kim, D. Frisch, and U. D. Hanebeck, “The ultimate hitchhiker’s guide to particle flow-based filters: A survey,” in <i>preparation</i>, 2026 - J6: S. Kim, M. Sun, I. Petrunin, and H.-S. Shin, “DBSCAN-based particle Gaussian mixture filters,” <i>Digital Signal Processing</i>, vol. 168, p. 105546, 2026 - J4: S. Kim, I. Petrunin, and H.-S. Shin, “AFJPDA: A multiclass multi-object tracking with appearance feature-aided joint probabilistic data association,” <i>Journal of Aerospace Information Systems</i>, vol. 21, no. 4, pp. 294–304, 2024 - J3: S. Kim, J. Kim, J. Park, and D. Lee, “Vision-based pose estimation of fixed-wing aircraft using you only look once and perspective-n-points,” <i>Journal of Aerospace Information Systems</i>, vol. 18, no. 9, pp. 659–664, 2021 - C3: S. Kim, S. Moon, I. Petrunin, H.-S. Shin, and S. Khattak, “Autonomous robotic radio source localization via a novel gaussian mixture filtering approach,” in <i>2025 28th International Conference on Information Fusion (FUSION)</i>, (Rio de Janeiro, Brazil), pp. 1–8, July 2025	
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LEADERSHIP AND TECHNICAL COMMUNICATION

Lecturer & Teaching Assistant (KIT & Cranfield University) Appointed to design and deliver a full-semester Master’s course on Localization of Mobile Agents (LMA) at KIT. Invited to deliver special guest lectures on State Estimation and Multi-target Tracking for Cranfield’s AVDC modules	
Technical Mentorship & Hardware Facilitation Supervised 2 Bachelor student theses focused on progressive filtering, localisation and multi-sensor fusion. Mentored postgraduate and undergraduate students during practical MATLAB sensor fusion implementations and embedded system hardware deployments	

HONOURS AND AWARDS

Honours - Fully-funded PhD Studentship, Cranfield University - Scholarship for Overseas Students, PNU Alumni Professors’ Association - Scholarship for Academic Excellence (full or partial)	September 2021–March 2025 July 2021 September 2015–February 2019
Awards 2022 ICNS Best Student Paper Award - KSAA Student Paper Competition - KSAS Student Paper Competition - Korea Robot Aircraft Competition - Drone Challenge Korea	2 nd Runner up in 2022 1 st Prize in 2020 1 st Prize in 2019 1 st Prize in 2018 and 3 rd Prize in 2019 3 rd Prize in 2018

SKILLS AND INTERESTS

Languages: Korean (Native/Fluent), English (Full Professional Proficiency), German (Basic)	
Applications: SolidWorks, Auto CAD, Microsoft Word/Excel/Visio and Adobe Photoshop	
Personal interests: Astronomy, photography and running (Marathon finisher: MK 2024 Seoul 2025 and Berlin 2026)	